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Consolidation Chemotherapy in postmolar low-risk gestational trophoblastic neoplasia: a systematic review protocol

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Consolidation Chemotherapy in postmolar low-risk gestational trophoblastic neoplasia: a systematic review protocol

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ABSTRACT

Introduction: Current evidence remains insufficient to strongly demonstrate the benefits of consolidation chemotherapy to all women with low-risk gestation trophoblastic neoplasia (GTN). This protocol outlines a systematic review to investigate whether consolidation chemotherapy is necessary for all patients with postmolar low-risk GTN after hCG normalization with first-line single-agent chemotherapy.

Methods and Analysis: A search string will be used to search PubMed (MEDLINE), EMBASE, Web of Sciences, Scopus, LILACS, and Cochrane Central Register of Controlled Trials (CENTRAL) databases. Articles will be screened using inclusion/exclusion criteria at the title and abstract level, and then at the full article level by two independent reviewers. Articles maintained after full article screening will undergo risk of bias assessment (GRADE system), and data will be extracted. The results will have important implications for women with low-risk GTN, as they might indicate that the burden of consolidation chemotherapy may be alleviated in some cases.

Ethics and dissemination: No formal ethical approval is required as all data collected will be secondary data and analyzed anonymously. Results will be disseminated through a peer-reviewed publication and at scientific events. No additional data are available

Systematic Review Registration: The protocol has been registered at the International Prospective Register of Systematic Reviews (PROSPERO; registration number: CRD42020164822)

Key words: low-risk gestational trophoblastic neoplasia, consolidation chemotherapy, recurrence

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Strengths and limitations of this study:

- This review will assess the relationship of patient, disease and treatment characteristics, as well relapse rates with the use of consolidation chemotherapy in low-risk GTN.
- The number of randomized controlled trials found is expected to be small as this type of study usually requires a large sample size, not easily available in the context of rare cancers such as GTN.
- The results may have implications for GTN research and management as they might indicate that the burden of treatment can be alleviated in some low-risk cases.

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INTRODUCTION

Gestational Trophoblastic Neoplasia (GTN) refers to a group of malignant lesions that arise from placental villous and extravillous trophoblast and may follow a hydatidiform mole or a nonmolar pregnancy (1,2) (Berkowitz & Goldstein, 2013; Berkowitz & Goldstein, 2019). Based on GTN classification and stage, treatment consists of either single agent therapy with methotrexate (MTX) or Actinomycin D for low-risk disease, or multi-agent therapy for high-risk disease (3)

Chemotherapy should be continued until complete remission is achieved, with a normalized serum human chorionic gonadotropin (hCG) level (4) . However, tumor resistance or relapse after first-line chemotherapy has been reported in approximately 25% of women with low-risk GTN (Lawrie et al., 2016). To eradicate the remaining tumor cells, consolidation therapy with the last effective agent(s) has been recommended to prevent relapse (4).

Few studies have addressed consolidation therapy for low-risk GTN. In 2006, Yang et al. identified less than two courses of consolidation chemotherapy as a risk factor for relapsed disease (6) while Sun et al. published results suggesting the opposite (7). This is likely to be due to the fact that besides using different definitions for GTN and relapse, those studies combined patients with postmolar GTN with patients with post-delivery GTN, and patients normalizing on MTX with patients who had progressive disease on MTX.

In 2012, Lybol et al. retrospectively compared relapse rates following two (The Netherlands) or three (UK) MTX consolidation courses in women completing MTX therapy for low-risk GTN. They found a 4 % relapse rate after 3 consolidation courses of MTX alternating with folinic acid in contrast to an 8.3 % relapse rate following 2

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consolidation courses, suggesting that 3 courses of consolidation chemotherapy is preferable to 2 in order to decrease the risk of low-risk GTN relapse. Nonetheless, differences between the Netherlands and the UK regarding the scoring systems employed in defining low-risk disease GTN, and in hCG measurement methods might have influenced the composition of the patient groups studied and the relapse rates observed (8).

In 2016, Couder et al. retrospectively analyzed the predictive factors of relapse in a homogeneous population with FIGO-2000-defined low risk GTN that received 2 consolidation courses after hCG normalization with MTX alone. They found a relapse rate of 5.7%, consistent with the rate reported in previous series where the FIGO scoring system was not uniformly used. Additionally, they demonstrated that postpartum low-risk GTN and patients who need more than 4 courses of MTX for normalization are at a higher risk of relapse than other low-risk patients (8.66-fold and 6.7-fold higher relapse risk, respectively). According to these authors, these findings suggest different biological behaviors among low-risk GTN patients and support previously reported relapse rates as low as 3% without any consolidation (9).

Taken together, these data indicate that current evidence on the benefits of consolidation chemotherapy to all women with low-risk GTN remains insufficiently strong. Thus, this review aims at investigating whether consolidation chemotherapy is necessary for all patients with postmolar low-risk GTN after hCG normalization with first-line single-agent chemotherapy. To this end, the proposed review will determine patient, disease and treatment characteristics of the women with postmolar low-risk GTN included in each of the studies of interest; and assess the relationship of relapse rate with these characteristics, use of consolidation chemotherapy, and number of consolidation chemotherapy cycles used (if any).

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Review Question

The components of population, intervention, comparator, outcome, and time frame (PICOT) were as follows:

Population: women with postmolar low-risk GTN who achieved remission (hCG normalization) with single-agent chemotherapy

Intervention: Consolidation chemotherapy after hCG normalization with first-line single-agent chemotherapy

Comparator: no consolidation treatment or placebo after hCG normalization with first-line single-agent chemotherapy.

Outcome: The primary outcome is rate of recurrence/relapse. Due to possible variation in disease definitions over time, definitions of outcomes will be extracted as reported in individual studies. Secondary outcomes will include consolidation chemotherapy toxicity and time to relapse/recurrence.

Time frame: up to one year of follow up after hCG normalization with first-line single-agent chemotherapy.

This protocol outlines the procedures for a systematic literature review intended to answer the question: Is consolidation chemotherapy necessary for all patients with postmolar low-risk gestational trophoblastic neoplasia to prevent recurrence after hCG normalization with first-line single-agent chemotherapy?

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METHODS

This protocol was designed according to PRISMA-P (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines (10) (checklist in supplemental file 1), and registered at the International Prospective Register of Systematic Reviews (PROSPERO; registration number: CRD42020164822).

Eligibility criteria

As information on consolidation therapy for postmolar low-risk gestational trophoblastic neoplasia is limited, the following randomized and non-randomized study designs will be included: randomized and quasi-randomized trials (including cluster and crossover trials), non-randomized controlled trials, prospective and retrospective cohort studies, case-control studies, controlled before-and-after studies, historically controlled studies, case series, cross-sectional studies, and interrupted time-series studies. Abstracts and conference proceedings will also be considered.

Case studies, commentaries, editorials, review articles, animal studies, and basic science studies, as well as studies not reporting relapse/recurrence rates and/or whether consolidation chemotherapy was delivered will be excluded. In studies also including women with other types of GTN (e.g. nonmolar, high risk), only data for the postmolar low-risk subgroup will be extracted. To ensure literature saturation, the reference lists of the studies included will be hand searched.

There will be no restrictions on date of publication, geographical location, study setting, or language of publication.

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Search strategy

The specific search strategies will be created by a Health Sciences Librarian with expertise in systematic review searching with input from the project team. Searches will be conducted in PubMed (MEDLINE), EMBASE, Web of Sciences, Scopus, LILACS, and Cochrane Central Register of Controlled Trials (CENTRAL). A draft PUBMED search strategy is presented in supplemental file 2. After the PUBMED strategy is finalized, it will be adapted to the syntax and subject headings of the other databases. Reference lists of included studies identified through the search will be scanned to identify any potentially eligible studies. The search will be updated toward the end of the review to capture recently published literature.

Data management

Two independent reviewers will perform data abstraction and quality assessment. Disagreements will be resolved through discussion and/or with the involvement of third-party arbitration. Using standardized forms, the reviewers will extract data independently and in duplicate from each eligible study. To ensure consistency across reviewers, calibration exercises will be conducted before starting the review. The Grading of Recommendations Assessment, Development and Evaluation Profiler (GRADEpro) software will be used to create tables for summary of findings and quality assessment.

Screening

All screening steps will be conducted independently by two reviewers. Duplicate references will be removed, and two review authors will independently screen the titles and abstracts yielded by the search. Full-text articles that meet the inclusion criteria will then be retrieved and screened. The reasons for excluding trials will be recorded.

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Neither of the review authors will be blind to the journal titles or to the study authors or institutions.

To facilitate collaboration among reviewers, search results will be uploaded to a web-based systematic review tool (e.g., Rayyan** or COVIDENCE) during the study selection. Training will be provided to members of the review team not familiar with such tools. Prior to the formal screening process, screening questions and forms will be developed and tested for assessments based on the inclusion and exclusion criteria.

The reviewers will meet after each stage to assess agreement. Disagreements will be resolved through discussion. If the conflict persists, a third independent reviewer will be consulted to reach unanimity.

The PRISMA flow diagram(10) will be used to report the screening process.

Data abstraction

A data abstraction form will be created and piloted by two reviewers on a sample subset of publications for this review. Data to be collected include study title, authors, publication date, number of patients with postmolar low-risk gestational trophoblastic neoplasia (GTN), low-risk GTN definition, type of antecedent pregnancy, pre-treatment hCG, single- agent chemotherapy regimen used, number of single-agent chemotherapy cycles needed to achieve hCG normalization, interval between chemotherapy cycles until hCG remission, number of consolidation chemotherapy cycles administered, consolidation chemotherapy adverse events, relapse/recurrence rate, relapse/recurrence definition, time to relapse/recurrence, relapse/recurrence prognostic factors identified.

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Quality assessment

The assessments of RCTs will be performed using the risk of bias tool from the Cochrane Collaboration (11), which evaluates potential bias for seven items across six domains: selection bias (random sequence generation; allocation concealment), performance bias (blinding of participants and personnel), detection bias (blinding of outcome assessment), attrition bias (incomplete outcome data), reporting bias (selective reporting), and other sources of bias. Each study will be rated as "high," "unclear" or "low" risk of bias

Non-randomized studies will be assessed using the Newcastle-Ottawa scale(12) that evaluates each study on eight items falling into three categories: (1) selection of study groups, (2) comparability of groups, and (3) ascertainment of exposure or outcome of interest.

The quality of evidence will be assessed using the Grading quality of evidence and strength of recommendations (GRADE) guidelines(13), which covers risk of bias, inconsistency, indirectness, imprecision, and publication bias. The overall quality of evidence is rated as high, moderate, low, or very low by each outcome measure.

The Grading of Recommendations Assessment, Development and Evaluation Profiler (GRADEpro) software will be used to create tables for summary of findings and quality assessment.(14)

Data analysis

Aggregate data will be used, and quantitative synthesis planned where possible if the studies included are sufficiently homogenous in terms of design and disease type or outcome definitions (Homogeneous outcomes in at least two studies). Continuous data will be expressed as means and standard deviation. For dichotomous data, the odds ratio

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will be calculated, and a log-rank approach will be used to estimate a hazard ratio, both with 95% CI. Random effects is the model chosen for meta-analysis as treatment effect is assumed to vary among studies. For quantitative synthesis, Review Manager 5.3 software will be used to pool the results of trials for each outcome.

Inconsistency among the results of the included studies will be ascertained by visual inspection of a forest plot (absence of overlap of CIs around the effect estimates of individual studies), as well as by the Higgins inconsistency test or I^2, where $I^2 > 50\ \%$ indicates moderate likelihood of heterogeneity.

The quality of evidence of intervention effect estimation for outcomes that can be plotted in meta-analysis will be generated according to GRADE (Grading of Recommendations Assessment, Development, and Evaluation).

Qualitative synthesis is planned for outcomes in which quantitative synthesis is not feasible or appropriate.

Any amendments to this protocol will be documented with their corresponding rationale in the full review.

DISCUSSION

Although chemotherapy is highly effective as the primary treatment for low-risk GTN -- with remission rates reaching up to 100% -- it can produce a range of adverse events and also have a considerable impact on quality of life. As GTN is a consequence of pregnancy, women and their partners must confront the loss of a pregnancy and are at risk of psychologic and social stress.

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The findings from this systematic review will have important implications for women with low-risk GTN, as they might indicate that the burden of the treatment may be alleviated in some cases.

Ethics and dissemination

Formal ethical approval is not required as all data collected will be secondary data and will be analyzed anonymously. The results will be disseminated through a peer-reviewed publication and at scientific conferences and meetings related to this field.

Data sharing statement

No additional data available.

Patient and public involvement (PPI)

There was no patient and public involvement (PPI) in this study.

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Conflict of interests

The authors declare no conflict of interest

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Author Contributions

MBS will serve as the first author of the protocol and review paper. She led all stages of protocol development, including development of the research question and objectives, search strategy, and extraction and analysis plans.

IM, KME, RSB, JFA, and NSH supervised and contributed to the plans for searching, screening, extracting, and writing phases. All authors read and approved the final manuscript.

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PubMed Search Strategy

("Low risk gestational trophoblastic neoplasia" OR "Recurrent trophoblastic disease" OR "Persistent trophoblastic disease" OR "hydatidiform mole" OR "Human chorionic gonadotropin" OR "Gestational trophoblastic tumor" OR "Persistent gestational trophoblastic tumor" OR "Trophoblastic Neoplasms" OR "Trophoblast Tumor" OR "Trophoblast Tumors" OR "Tumor, Trophoblast" OR "Tumors, Trophoblast" OR "Neoplasms, Trophoblastic" OR "Neoplasm, Trophoblastic" OR "Trophoblastic Neoplasm" OR "Trophoblast Neoplasms" OR "Neoplasm, Trophoblast" OR "Neoplasms, Trophoblast" OR "Trophoblast Neoplasm" OR "Trophoblastic Tumor" OR "Trophoblastic Tumors" OR "Tumor, Trophoblastic" OR " Tumors, Trophoblastic" OR "Trophoblastic Cancer" OR "Cancer, Trophoblastic" OR "Cancers, Trophoblastic" OR "Trophoblastic Cancers" OR "Trophoblast Cancer" OR "Cancer, Trophoblast" OR "Cancers, Trophoblast" OR "Trophoblast Cancers" OR "Disease, Gestational Trophoblastic" OR "Diseases, Gestational Trophoblastic" OR "Gestational Trophoblastic Diseases" OR "Trophoblastic Disease, Gestational" OR "Trophoblastic Diseases, Gestational" OR "Gestational Trophoblastic Neoplasia" OR "Neoplasia, Gestational Trophoblastic" OR "Trophoblastic Neoplasia, Gestational" OR "Trophoblastic Neoplasms, Gestational" OR "Gestational Trophoblastic Neoplasms" OR "Gestational Trophoblastic Neoplasm" OR "Neoplasm, Gestational Trophoblastic" OR "Neoplasms, Gestational Trophoblastic" OR "Trophoblastic Neoplasm, Gestational") AND ("consolidation course" OR "Consolidation Chemotherapy" OR "Chemotherapies, Consolidation" OR "Chemotherapy, Consolidation" OR "Consolidation Chemotherapies" OR "methotrexate, toxicity" OR "actinomycin" OR "methotrexate, relapse" OR Methotrexate) AND (Recurrence OR Recurrences OR Recrudescence OR Recrudescences OR Relapse OR Relapses)

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PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist: recommended items to address in a systematic review protocol*

Section and topic	Item No	Checklist item	Reported item
ADMINISTRATIVE INFORMATION			
Title:			Page 1
Identification	1a	Identify the report as a protocol of a systematic review	
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	N/A
Registration	2	If registered, provide the name of the registry (such as PROSPERO) and registration number	Page 2&7
Authors:			
Contact	3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	Page 1
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	Page 13
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	N/A
Support:			
Sources	5a	Indicate sources of financial or other support for the review	N/A
Sponsor	5b	Provide name for the review funder and/or sponsor	N/A
Role of sponsor or funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	N/A
INTRODUCTION			
Rationale	6	Describe the rationale for the review in the context of what is already known	Pages 4&5
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	Page 6
METHODS			
Eligibility criteria	8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	Page 6 &7
Information sources	9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	Page 7
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	Supp. File 2
Study records:			
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	Page 8
Selection process	11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	Page 8&9
Data collection process	11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	Page 9
Data items	12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	Page 9

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Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	Page 9
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	Page 10
Data synthesis	15a	Describe criteria under which study data will be quantitatively synthesised	Page 10 & 11
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I^2, Kendall's τ)	Page 11
	15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	N/A
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	Page 11
Confidence in	17	Describe how the strength of the cumulative evidence body of evidence will be assessed (such as GRADE)	Page 11

From: Shamseer L, Moher D, Clarke M, Ghersi D, Liberati A, Petticrew M, Shekelle P, Stewart L, PRISMA-P Group. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015: elaboration and explanation. BMJ. 2015 Jan 2;349(jan02 1):g7647.

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ABSTRACT

Introduction: Current evidence remains insufficient to strongly demonstrate the benefits of consolidation chemotherapy to all women with low-risk gestational trophoblastic neoplasia (GTN). This protocol outlines a systematic review to investigate whether consolidation chemotherapy is necessary for all patients with postmolar low-risk GTN after hCG normalization with first-line single-agent chemotherapy.

Methods and analysis: A search string will be used to search the PubMed (MEDLINE), EMBASE, Web of Sciences, Scopus, LILACS, and Cochrane Central Register of Controlled Trials (CENTRAL) databases. Articles will be screened at the title and abstract level, and then at the full article level by two independent reviewers using inclusion/exclusion criteria. Randomized and non-randomized study designs will be included, while case studies, commentaries, editorials, review articles, animal studies, basic science studies, and cross-sectional studies, as well as studies not reporting relapse/recurrence rates and/or whether consolidation chemotherapy was delivered will be excluded. There will be no restrictions on date of publication, geographical location, study setting, or language of publication. The primary outcome is rate of recurrence/relapse. The assessments of randomized controlled trials will be performed using the risk of bias tool from the Cochrane Collaboration. Non-randomized studies will be assessed using the Newcastle-Ottawa scale. The quality of evidence will be assessed using the Grading quality of evidence and strength of recommendations (GRADE) guidelines.

Ethics and dissemination: No formal ethical approval is required as all data collected will be secondary data and analyzed anonymously. Results will be disseminated through a peer-reviewed publication and at scientific events.

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Systematic Review Registration: The protocol has been registered at the International Prospective Register of Systematic Reviews (PROSPERO; registration number: CRD42020164822).

Strengths and limitations of this study

- The number of randomized controlled trials found is expected to be small as this type of study usually requires a large sample size not easily available in the context of rare cancers such as GTN.
- Differences in GTN diagnostic and classification criteria, as well as in the definition of relapse, may confound comparisons
- The rigorous and transparent study design will possibly reduce the risk of biases.
- Grading the quality of the evidence will provide confidence in the effect estimates

INTRODUCTION

Gestational Trophoblastic Neoplasia (GTN) refers to a group of malignant lesions that arise from placental villous and extravillous trophoblast and may follow a hydatidiform

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mole or a nonmolar pregnancy (1,2). Based on GTN classification and stage, treatment consists of either single agent therapy with methotrexate (MTX) or Actinomycin D for low-risk disease, or multi-agent therapy for high-risk disease (3).

Chemotherapy should be continued until complete remission is achieved, with a normalized serum human chorionic gonadotropin (hCG) level (4) . However, tumor resistance or relapse after first-line chemotherapy has been reported in approximately 25% of women with low-risk GTN(5). To eradicate the remaining tumor cells, consolidation therapy with the last effective agent(s) has been recommended to prevent relapse (4).

Few studies have addressed consolidation therapy for low-risk GTN. In 2006, Yang et al. identified less than two courses of consolidation chemotherapy as a risk factor for relapsed disease (6)while Sun et al. published results suggesting the opposite (7). This is likely to be due to the fact that besides using different definitions for GTN and relapse, those studies combined patients with postmolar GTN with patients with post-delivery GTN, and patients normalizing on MTX with patients who had progressive disease on MTX.

In 2012, Lybol et al. retrospectively compared relapse rates following two (The Netherlands) or three (UK) MTX consolidation courses in women completing MTX therapy for low-risk GTN. They found a 4 % relapse rate after three consolidation courses of MTX alternating with folinic acid in contrast to an 8.3 % relapse rate following two consolidation courses, suggesting that three courses of consolidation chemotherapy is preferable to two in order to decrease the risk of low-risk GTN relapse. Nonetheless, differences between the Netherlands and the UK regarding the scoring systems employed in defining low-risk disease GTN, and in hCG measurement methods

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might have influenced the composition of the patient groups studied and the relapse rates observed(8).

In 2016, Couder et al. retrospectively analyzed the predictive factors of relapse in a homogeneous population with FIGO-2000-defined low risk GTN that received two consolidation courses after hCG normalization with MTX alone. They found a relapse rate of 5.7%, consistent with the rate reported in previous series where the FIGO scoring system was not uniformly used. Additionally, they demonstrated that postpartum low-risk GTN and patients who need more than four courses of MTX for normalization are at a higher risk of relapse than other low-risk patients (8.66-fold and 6.7-fold higher relapse risk, respectively). According to these authors, these findings suggest different biological behaviors among low-risk GTN patients and support previously reported relapse rates as low as 3% without any consolidation(9).

Taken together, these data indicate that current evidence on the benefits of consolidation chemotherapy to all women with low-risk GTN remains insufficiently strong. Thus, this review aims at investigating whether consolidation chemotherapy is necessary for all patients with postmolar low-risk GTN after hCG normalization with first-line single-agent chemotherapy. To this end, the proposed review will determine patient, disease and treatment characteristics of the women with postmolar low-risk GTN included in each of the studies of interest; and assess the relationship of relapse rate with these characteristics, use of consolidation chemotherapy, and number of consolidation chemotherapy cycles used (if any).

Review Question

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The components of population, intervention, comparator, outcome, and time frame (PICOT) are as follows:

Population: women with postmolar low-risk GTN who achieved remission (hCG normalization) with single-agent chemotherapy

Intervention: Consolidation chemotherapy after hCG normalization with first-line single-agent chemotherapy

Comparator: No consolidation treatment or placebo after hCG normalization with first-line single-agent chemotherapy.

Outcome: The primary outcome is rate of recurrence/relapse. Due to possible variation in disease definitions over time, definitions of outcomes will be extracted as reported in individual studies. Secondary outcomes will include consolidation chemotherapy toxicity and time to relapse/recurrence.

Time frame: up to one year of follow up after hCG normalization with first-line single-agent chemotherapy.

This protocol outlines the procedures for a systematic literature review intended to answer the question: Is consolidation chemotherapy necessary for all patients with postmolar low-risk gestational trophoblastic neoplasia to prevent recurrence after hCG normalization with first-line single-agent chemotherapy?

METHODS AND ANALYSIS

This protocol was designed according to PRISMA-P (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines (10)(checklist in supplemental file 1) and registered at the International Prospective Register of Systematic Reviews (PROSPERO; registration number: CRD42020164822).

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Eligibility criteria

As information on consolidation therapy for postmolar low-risk gestational trophoblastic neoplasia is limited, the following randomized and non-randomized study designs will be included: randomized and quasi-randomized trials (including cluster and crossover trials), non-randomized controlled trials, prospective and retrospective cohort studies, case-control studies, controlled before-and-after studies, historically controlled studies, case series, and interrupted time-series studies. Abstracts and conference proceedings will also be considered.

Case studies, commentaries, editorials, review articles, animal studies, and basic science studies, cross sectional studies, as well as studies not reporting relapse/recurrence rates and/or whether consolidation chemotherapy was delivered will be excluded. In studies also including women with other types of GTN (e.g. nonmolar, high risk), only data for the postmolar low-risk subgroup will be extracted. To ensure literature saturation, the reference lists of the studies included will be hand searched.

There will be no restrictions on date of publication, geographical location, study setting, or language of publication.

Search strategy

The specific search strategies will be created by a Health Sciences Librarian with expertise in systematic review searching with input from the project team. Searches will be conducted in PubMed (MEDLINE), EMBASE, Web of Sciences, Scopus, LILACS, and Cochrane Central Register of Controlled Trials (CENTRAL). A draft search strategy for all databases to be used is presented in supplemental file 2. After the PUBMED strategy is finalized, it will be adapted to the syntax and subject headings of the other databases. Reference lists of included studies identified through the search will

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be scanned to identify any potentially eligible studies. The search will be updated toward the end of the review to capture recently published literature.

Data management

Two independent reviewers will perform data abstraction and quality assessment. Disagreements will be resolved through discussion and/or with the involvement of third-party arbitration. Using standardized forms, the reviewers will extract data independently and in duplicate from each eligible study. To ensure consistency across reviewers, calibration exercises will be conducted before starting the review. The Grading of Recommendations Assessment, Development and Evaluation Profiler (GRADEpro) software will be used to create tables for summary of findings and quality assessment.

Screening

All screening steps will be conducted independently by two reviewers. Duplicate references will be removed, and two review authors will independently screen the titles and abstracts yielded by the search. Full-text articles that meet the inclusion criteria will then be retrieved and screened. The reasons for excluding trials will be recorded. Neither of the review authors will be blind to the journal titles or to the study authors or institutions.

To facilitate collaboration among reviewers, search results will be uploaded to a web-based systematic review tool (e.g., Rayyan** or COVIDENCE) during the study selection. Training will be provided to members of the review team not familiar with such tools. Prior to the formal screening process, screening questions and forms will be developed and tested for assessments based on the inclusion and exclusion criteria.

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The reviewers will meet after each stage to assess agreement. Disagreements will be resolved through discussion. If the conflict persists, a third independent reviewer will be consulted to reach unanimity.

The PRISMA flow diagram (10) will be used to report the screening process.

Data abstraction

A data abstraction form will be created and piloted by two reviewers on a sample subset of publications for this review. Data to be collected include study title, authors, publication date, number of patients with postmolar low-risk gestational trophoblastic neoplasia (GTN), low-risk GTN definition, type of antecedent pregnancy, pre-treatment hCG, single- agent chemotherapy regimen used, number of single-agent chemotherapy cycles needed to achieve hCG normalization, interval between chemotherapy cycles until hCG remission, number of consolidation chemotherapy cycles administered, consolidation chemotherapy adverse events, relapse/recurrence rate, relapse/recurrence definition, time to relapse/recurrence, relapse/recurrence prognostic factors identified.

Quality assessment

The assessments of RCTs will be performed using the risk of bias tool from the Cochrane Collaboration (11), which evaluates potential bias for seven items across six domains: selection bias (random sequence generation; allocation concealment), performance bias (blinding of participants and personnel), detection bias (blinding of outcome assessment), attrition bias (incomplete outcome data), reporting bias (selective reporting), and other sources of bias. Each study will be rated as “high,” “unclear” or “low” risk of bias.

Non-randomized studies will be assessed using the Newcastle-Ottawa scale (12)that evaluates each study on eight items falling into three categories: (1) selection of study

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groups, (2) comparability of groups, and (3) ascertainment of exposure or outcome of interest.

The quality of evidence will be assessed using the Grading quality of evidence and strength of recommendations (GRADE) guidelines (13) which covers risk of bias, inconsistency, indirectness, imprecision, and publication bias. The overall quality of evidence is rated as high, moderate, low, or very low by each outcome measure.

The Grading of Recommendations Assessment, Development and Evaluation Profiler (GRADEpro) software will be used to create tables for summary of findings and quality assessment. (14)

Data analysis

Aggregate data will be used, and quantitative synthesis planned where possible if the studies included are sufficiently homogenous in terms of design and disease type or outcome definitions (Homogeneous outcomes in at least two studies). Continuous data will be expressed as means and standard deviation. For dichotomous data, the odds ratio will be calculated, and a log-rank approach will be used to estimate a hazard ratio, both with 95% CI. Random effects is the model chosen for meta-analysis as treatment effect is assumed to vary among studies. For quantitative synthesis, Review Manager 5.3 software will be used to pool the results of trials for each outcome.

Inconsistency among the results of the included studies will be ascertained by visual inspection of a forest plot (absence of overlap of CIs around the effect estimates of individual studies), as well as by the Higgins inconsistency test or I^2, where $I^2 > 50\ \%$ indicates moderate likelihood of heterogeneity.

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The quality of evidence of intervention effect estimation for outcomes that can be plotted in meta-analysis will be generated according to GRADE (Grading of Recommendations Assessment, Development, and Evaluation).

Qualitative synthesis is planned for outcomes in which quantitative synthesis is not feasible or appropriate.

Any amendments to this protocol will be documented with their corresponding rationale in the full review.

Patient and public involvement

There is no patient and public involvement in this study.

ETHICS AND DISSEMINATION

Formal ethical approval is not required as all data collected will be secondary data and will be analyzed anonymously. The results will be disseminated through a peer-reviewed publication and at scientific conferences and meetings related to this field.

Acknowledgments

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Conflict of interests

The authors declare no conflicts of interest.

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Contributors

MBS will serve as the first author of the protocol and review paper. She led all stages of protocol development, including development of the research question and objectives, search strategy, and extraction and analysis plans. IM, KME, RSB, JFA, and NSH supervised and contributed to the plans for searching, screening, extracting, and writing phases. All authors read and approved the final manuscript.

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PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist: recommended items to address in a systematic review protocol*

Section and topic	Item No	Checklist item	Reported item
ADMINISTRATIVE INFORMATION			
Title:			Page 1
Identification	1a	Identify the report as a protocol of a systematic review	
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	N/A
Registration	2	If registered, provide the name of the registry (such as PROSPERO) and registration number	Page 2&7
Authors:			
Contact	3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	Page 1
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	Page 13
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	N/A
Support:			
Sources	5a	Indicate sources of financial or other support for the review	N/A
Sponsor	5b	Provide name for the review funder and/or sponsor	N/A
Role of sponsor or funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	N/A
INTRODUCTION			
Rationale	6	Describe the rationale for the review in the context of what is already known	Pages 4&5
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	Page 6
METHODS			
Eligibility criteria	8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	Page 6 &7
Information sources	9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	Page 7
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	Supp. File 2
Study records:			
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	Page 8
Selection process	11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	Page 8&9
Data collection process	11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	Page 9
Data items	12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	Page 9

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Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	Page 9
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	Page 10
Data synthesis	15a	Describe criteria under which study data will be quantitatively synthesised	Page 10 & 11
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I^2, Kendall's τ)	Page 11
	15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	N/A
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	Page 11
Confidence in	17	Describe how the strength of the cumulative evidence body of evidence will be assessed (such as GRADE)	Page 11

From: Shamseer L, Moher D, Clarke M, Ghersi D, Liberati A, Petticrew M, Shekelle P, Stewart L, PRISMA-P Group. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015: elaboration and explanation. BMJ. 2015 Jan 2;349(jan02 1):g7647.

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SEARCH STRATEGIES

Cochrane Central

*low risk gestational trophoblastic neoplasia in All Text AND recurrence OR relapse in All Text

EMBASE

low AND (risk/exp OR risk) AND ("gestational trophoblastic disease"/exp OR (gestational AND trophoblastic AND disease) OR "gestational trophoblastic disease" OR (gestational AND trophoblastic AND neoplasia) OR "gestational trophoblastic neoplasia") AND (recurrance OR recurrence/exp OR recurrence OR recurrences OR recurrencies OR recurrency OR recurrent OR recurrently OR recurrents OR (recurrence/exp OR recurrence OR relapse OR relapses OR relapsing OR relapsed OR relapser OR relapsers))

LILACS

(Trophoblastic Neoplasms or Neoplasias Trofoblásticas or Câncer Trofoblástico or Tumor Trofoblástico or MH: C04.557.465.955 or C04.850.908 or C13.703.720.949 or Gestational Trophoblastic Disease or Enfermedad Trofoblástica Gestacional or Doença Trofoblástica Gestacional or Neoplasias Trofoblásticas Gestacionais or MH: C04.557.465.955.416 or MH:C04.850.908.416 or MH:C13.703.720.949.416 or Neoplasia Trofoblástica Gestacional de baixo risco or Neoplasma Trofoblástico Gestacional de baixo risco) [Palavras] and recorrência OR recidiva OR recurrence OR relapse OR recorrencia OR récurrence [Palavras]

PUBMED

"low"[All Fields] AND ("risk"[MeSH Terms] OR "risk"[All Fields]) AND ("gestational trophoblastic disease"[MeSH Terms] OR ("gestational"[All Fields] AND "trophoblastic"[All Fields] AND "disease"[All Fields]) OR "gestational trophoblastic disease"[All Fields] OR ("gestational"[All Fields] AND "trophoblastic"[All Fields] AND "neoplasia"[All Fields]) OR "gestational trophoblastic neoplasia"[All Fields]) AND ("recurrance"[All Fields] OR "recurrence"[MeSH Terms] OR "recurrence"[All Fields] OR "recurrences"[All Fields] OR "recurrencies"[All Fields] OR "recurrency"[All Fields] OR "recurrent"[All Fields] OR "recurrently"[All Fields] OR "recurrents"[All Fields] OR ("recurrence"[MeSH Terms] OR "recurrence"[All Fields] OR "relapse"[All Fields] OR "relapses"[All Fields] OR "relapsing"[All Fields] OR "relapsed"[All Fields] OR "relapser"[All Fields] OR "relapsers"[All Fields]))

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recurrence OR relapse OR relapses OR relapsing OR relapsed OR relapser OR relapsers)
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Web of sciences

ALL=(low AND (risk OR risk) AND ("gestational trophoblastic disease" OR (gestational AND trophoblastic AND disease) OR "gestational trophoblastic disease" OR (gestational AND trophoblastic AND neoplasia) OR "gestational trophoblastic neoplasia") AND (recurrance OR recurrence OR recurrence OR recurrences OR recurrencies OR recurrency OR recurrent OR recurrently OR recurrents OR (recurrence OR recurrence OR relapse OR relapses OR relapsing OR relapsed OR relapser OR relapsers)))

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